R2/R squared

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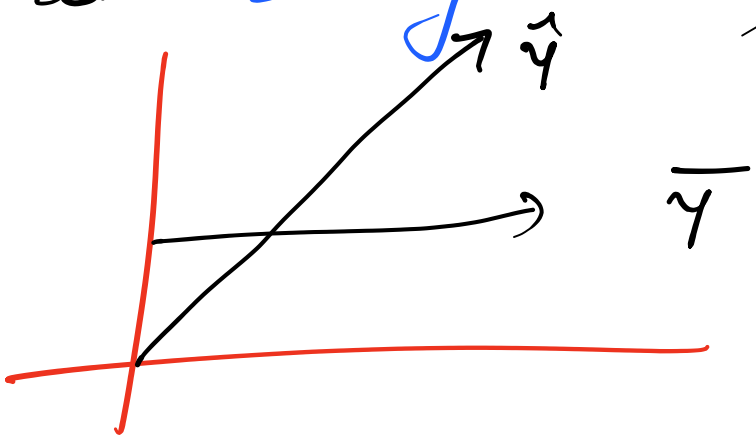
R^2 Score

* It is used to measure how good is a model.
* R^2 score measures how much variance explained by the model.
* R^2 score compares Regression line ($\hat{y}$) with average line ($\bar{y}$).



$$R^2 = 1 - \frac{RSS}{TSS}$$

$$\boxed{R^2 = 1 - \frac{\sum (y - \hat{y})^2}{\sum (y - \bar{y})^2}}$$

$R^2 \longrightarrow 0$ to 1

$R^2 \longrightarrow 1$ — Good model

R^2 is near to 1 — Good model.

R^2 is near to 0 — Bad model.

RSS → Residual Sum of Squares

Error made by regression line

$$RSS = \frac{1}{n}\sum(y-\hat{y})^2$$

TSS → Total Sum of Squares

Error made by average line

$$TSS = \frac{1}{n}\sum(y-\bar{y})^2.$$

Limitations

As number of irrelevant input features increases the R^2 score also increases.

Adjusted R^2 Score

→ It is used to overcome limitation of R^2 Score.

→ Adjusted R^2 Score will [illegible] added variables

Check will having
Contribute to model.

→ Adjusted R^2 < R^2 Score.

$$adj = 1 - \frac{(1-R^2)(N-1)}{(N-P-1)}$$

$R^2 \rightarrow R^2$ Score

$N \rightarrow$ No of observations

$P \rightarrow$ No of input variables.